

The Definition of Fisher Information

- We give a lower bound on the variance of any unbiased estimator.
- We show that, under regularity conditions, the variances of the maximum likelihood estimates achieve this lower bound asymptotically.
- Additional Regularity Conditions:
 - R3: The pdf $f(x; \theta)$ is twice differentiable as a function of θ .
 - R4: The integral $\int f(x; \theta) dx$ can be differentiated twice under the integral sign as a function of θ .

$$1 = \int_{-\infty}^{\infty} f(x; \theta) dx$$

$$0 = \int_{-\infty}^{\infty} \frac{\partial f(x; \theta)}{\partial \theta} dx$$

$$0 = \int_{-\infty}^{\infty} \frac{\partial f(x; \theta) / \partial \theta}{f(x; \theta)} f(x; \theta) dx$$

$$0 = \int_{-\infty}^{\infty} \frac{\partial \log f(x; \theta)}{\partial \theta} f(x; \theta) dx$$

$$\mathbb{E}\left[\frac{\partial \log f(x; \theta)}{\partial \theta}\right] = 0$$

$$0 = \int_{-\infty}^{\infty} \frac{\partial^2 \log f(x; \theta)}{\partial \theta^2} f(x; \theta) dx + \int_{-\infty}^{\infty} \frac{\partial \log f(x; \theta)}{\partial \theta} \frac{\partial \log f(x; \theta)}{\partial \theta} f(x; \theta) dx$$

$$\mathbb{E}\left[\left(\frac{\partial \log f(x; \theta)}{\partial \theta}\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2 \log f(x; \theta)}{\partial \theta^2}\right]$$

- **Fisher Information** $I(\theta)$:

$$I(\theta) = \int_{-\infty}^{\infty} \frac{\partial \log f(x; \theta)}{\partial \theta} \frac{\partial \log f(x; \theta)}{\partial \theta} f(x; \theta) dx = \mathbb{E} \left[\left(\frac{\partial \log f(x; \theta)}{\partial \theta} \right)^2 \right]$$

$$I(\theta) = - \int_{-\infty}^{\infty} \frac{\partial^2 \log f(x; \theta)}{\partial \theta^2} f(x; \theta) dx = -\mathbb{E} \left[\frac{\partial^2 \log f(x; \theta)}{\partial \theta^2} \right]$$

$$I(\theta) = \text{Var} \left(\frac{\partial \log f(x; \theta)}{\partial \theta} \right)$$

- **Score Function:**

$$\frac{\partial \log f(x; \theta)}{\partial \theta}$$

Recall that it determines the estimating equations for the mle; that is; the mle $\hat{\theta}$ solves

$$\sum_{i=1}^n \frac{\partial \log f(x_i; \theta)}{\partial \theta} = 0$$

for θ .

- Information for a Bernoulli Random Variable: Let X be Bernoulli $b(1; \theta)$.
Thus,

$$\log f(x; \theta) = x \log \theta + (1 - x) \log(1 - \theta),$$

$$\frac{\partial \log f(x; \theta)}{\partial \theta} = \frac{x}{\theta} - \frac{1 - x}{1 - \theta},$$

$$\frac{\partial^2 \log f(x; \theta)}{\partial \theta^2} = -\frac{x}{\theta^2} - \frac{1 - x}{(1 - \theta)^2}.$$

$$I(\theta) = -\mathbb{E}\left[-\frac{x}{\theta^2} - \frac{1 - x}{(1 - \theta)^2}\right] = \frac{1}{\theta(1 - \theta)}$$

- Information for a Location Family: Consider a random sample $X_1, \dots, X_n$ such that

$$X_i = \theta + e_i, \quad i = 1, \dots, n,$$

where $e_1, \dots, e_n$ are iid with common pdf $f(x)$. Then the common pdf of X_i is $f_X(x; \theta) = f(x - \theta)$. This model is called a *location model*. Then the information is

$$\begin{aligned} I(\theta) &= \int_{-\infty}^{\infty} \left(\frac{f'(x - \theta)}{f(x - \theta)} \right)^2 f(x - \theta) dx \\ &= \int_{-\infty}^{\infty} \left(\frac{f'(x)}{f(x)} \right)^2 f(x) dx. \end{aligned}$$

The information does not depend on θ .

- Suppose that X_i has the Laplace pdf, $f(x; \theta) = \frac{1}{2}e^{-|x-\theta|}$. Then it is easy to see that we can write X_i as

$$X_i = \theta + e_i,$$

where $e_1, \dots, e_n$ are iid from pdf $f(z) = 2^{-1} \exp(-|z|)$.

$$f'(z) = -2^{-1} \operatorname{sgn}(z) \exp(-|z|)$$

$$I(\theta) = \int_{-\infty}^{\infty} \left(\frac{f'(x)}{f(x)} \right)^2 f(x) dx = 1.$$